

Co-founder and Vice-President, Aurora Rocketry

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1 Introduction: Aurora Rocketry

Aurora Rocketry is a student association of the University of Bologna. I founded it together with the first President, Nicolò Basso, in March 2024, at the Industrial Engineering site of the Forlì campus. By July 2025 it had more than 100 students from several faculties.

The association designs and builds model rockets for the annual EuRoC competition (European Rocketry Challenge), organised by the Portuguese Space Agency near Constância (PT). During the event, several European student teams launch their prototypes, which are evaluated on design, flight performance and teamwork. The goal of the association is to take part in this competition and to obtain the best possible result.

Aurora Rocketry therefore has an educational purpose, such as developing technical and organisational skills. Each department deals with specific aspects of the design and requires specific skills. Based on their own experience, interests and inclinations, each student chooses the department that suits them best and contributes to its activities, which, together with those of the other departments, lead to the common objectives.

In this way Aurora Rocketry gives students a real opportunity to get closer to the world of engineering, by working on a tangible project already during their academic path.

One year and a half after its foundation, the team was admitted to the 2025 edition of EuRoC and made its debut at the competition with the prototype *Nemesis*, finishing second in the 3000 m COTS category and eighth in the overall ranking.

2 Administrative activities

The administrative activities described below are presented in chronological order and show my work in the related context, from the foundation of the association to our participation in EuRoC 2025.

2.1 Foundation

Although the association was officially founded in March 2024, with the first public presentation and the admission of the first members, the idea came up in September and October 2023.

Those months were needed to develop the project in a consistent and structured way, so that it could later be presented in an informed manner, showing credibility, competence and seriousness. In that period, Basso and I defined our tangible objective: taking part in EuRoC. Everything else was built around this cornerstone.

Several meetings with teams that were already active (for example Skyward of Politecnico di Milano) and with some professors, including Fabrizio Ponti and Nabil Souhair, turned out to be essential. I prepared a wide set of questions for the people we met, mainly about:

- advice on how to manage human resources, technical activities and funds in a responsible way;
- how to recruit new members and how to present the project properly to the first candidates and to potential sponsors;
- how to handle risks, responsibilities and legal permits, which are unavoidable in this kind of activity.

These meetings allowed us to start the project in a more informed and aware way, and they clarified several doubts. From this experience I learned how important it is to set pride aside and to make use of the knowledge of institutions and of more experienced people: this kind of exchange helps to solve known problems and to discover solutions to issues that we had not considered yet.

After that, I studied the competition rules in detail. It is interesting how a document that sets terms and conditions on the design and construction of model rockets becomes, for a student who approaches the field for the first time, the most reliable guide to building them. Reading it gives a solid basis from which the single parts can be studied in more depth.

The activities that emerged from this preliminary study were then divided into main groups, which form the current department structure of the association.

After writing the operational plan, including the work profile of each department and the list of its activities, we started the first recruitment phase, through university presentations and the distribution of flyers, some of which I designed myself.

In this phase every candidate could freely choose a department and join without any condition. All of them were admitted. This decision had already been taken before the recruitment started, both because we did not expect a large number of applications, and because we did not yet feel entitled to decide whether a student should be admitted or not.

The candidates who showed the strongest interest, reliability and passion, through a dedicated Google form, were selected as Department Heads. I defined the duties and

responsibilities of the Department Heads, and then wrote them down in an internal regulation.

Once the first selection was over, I partly managed the introductory technical meetings with the Department Heads. The main objectives were:

- introducing the technical guidelines to follow and the general design of the prototype, based on what we had learned in the previous months;
- promoting an iterative approach (which is still used today);
- helping the Department Heads to grow into their roles step by step, so that they could manage the activities of their own departments independently.

At this point the technical management was assigned to people with specific responsibilities, and the design of the first prototype started: *Borealis*.

Borealis was deliberately simple. It was meant as a first hands-on experience, useful to observe how the work was distributed and to bring design and manufacturing problems to the surface. It was not intended for EuRoC, and it was designed to reach about 1500 m, while the lowest EuRoC category requires at least 3000 m.

During this foundation phase and the start of the *Borealis* project, I mainly dealt with administrative and managerial activities. During the development of the *Nemesis* project, instead, I also worked on technical activities, mainly related to CFD analyses and structural checks of the prototype. All these activities are described in this report.

2.2 Legal registration and registration in the university register

In parallel with the start of *Borealis*, I took care of the administrative steps needed to access the funding and the opportunities offered by the university and by industry.

The first step was the formal registration of the association with the Italian Revenue Agency as a university student association, followed by the enrolment in the register of associations of the University of Bologna.

The process started with the drafting of the statutory documents (deed of foundation and statute). Several weeks were dedicated to this phase: the documents had to meet the requirements of the regulation of the register, which are quite strict. My main mistake was to customise the statute too much, which created conflicts with the university regulations, especially regarding the Assembly, its powers and the election procedures. After several attempts, all of them rejected, we decided to adopt the template provided by the university and to fill in only the necessary fields. The lesson I learned is that, in bureaucratic matters, it is better to follow the university templates and to avoid unnecessary customisation.

This activity was followed by the legal registration with the Revenue Agency. Since clear sources were hard to find, I asked for support from Silvia Poggi, a lawyer, and organised a dedicated meeting with her. The clarifications she gave, combined with the scattered information found on several websites, allowed us to complete the following steps:

- creation of the tax code at the Revenue Agency;
- payment of the registration fee at the post office;
- return to the Agency for the final registration.

Each step required the completion of the forms F24, AA5/6 and MOD. 69.

In about one month the association was registered, so we moved on to the enrolment in the university register. We had to prove that we had reached 100 members, all of them meeting the requirements of the University of Bologna, which meant collecting paper signatures and personal data. Reaching all of them required careful management of my schedule, of my contacts and of a few delegated tasks.

Within a few weeks this activity was also completed, and it started to produce the expected results.

2.3 Facilities and first funding

Once the first phase was over, we could start looking for funding. Being a new team, with no previous projects, made it harder to attract sponsors compared to an association with tangible results.

However, Aurora Rocketry was not the only young group. The *THRUST Team* of the University of Padua was also at its beginning, but it had already found funds for its first prototype. What they were missing was the legal registration process, and they were looking for practical guidance on how to proceed. Aurora Rocketry could offer it. I therefore organised a meeting for a fair exchange of ideas: they would introduce us to the world of sponsorship, and I would explain the registration process.

Their advice, which was reasonable, was to contact a large number of companies, trusting that the law of large numbers would produce some positive answers. I prepared a standard email and sent it to many companies in the area, but the result was almost nil. It was clear that the method had to be corrected.

Analysing the problem, we understood that the main mistake was the length of the emails: walls of text full of explanations, motivations and direct requests for support did not work. A short message, with an invitation to an introductory online meeting, would have been much more effective.

At that moment, while the search for sponsors was giving discouraging results, the association came across a call promoted by the Emilia-Romagna Region in collaboration with the European Union. It provided for the distribution of NRRP funds to organisations and associations with extracurricular projects.

It was a unique opportunity that we could not miss. We found out about it only a few days before the deadline, so preparing the application, collecting data and writing a financial plan became a race against time. But we managed to do it.

A few weeks later the outcome arrived: 20,000 € allocated to our project. The impact on the association was huge: relief, enthusiasm and the feeling that a significant part of our problems had been solved. Three major difficulties, however, remained:

- **Access to the funds.** This was not immediate: buying materials or components required at least two quotations from different companies, as required by the regulations, in order to prove that a market survey had been carried out. This system is certainly not well suited to our operational needs, but it is still manageable.
- **Working spaces.** The MaSTeR Lab laboratory in via Cicognani was the ideal solution. After a meeting with Paolo Proli (technical unit manager) and Fabio Ramilli, we were given a workbench and a dedicated space for materials and projects.
- **Launch permits.** Already during the first meeting with Skyward, the bureaucratic complexity of launches in Italy became clear: a *NOTAM* (Notice To AirMen)

was needed, a document issued by ENAV that notifies the temporary use of a portion of airspace. For safety reasons, requesting a NOTAM means proving the reliability of the activity through simulations, data and technical documentation. The process can take months, so we decided to start immediately. At the same time, some other students and I were attending the Air Traffic Control course of professor Fabrizio Olivetti, who is responsible for the ENAV Training Centre of Forlì. He was helpful from the start and provided a document describing several potential launch sites, some of which were already covered by permanent NOTAMs thanks to the frequent local air activity.

Despite this, as will be explained later, the *Borealis* prototype was never launched.

2.4 Opening of the association bank account

This phase was followed by the opening of the association bank account at LaBCC. This choice turned out to be efficient, since LaBCC became one of the first official sponsors of Aurora Rocketry, contributing with a significant amount of funding.

2.5 Internal regulation

In the following period we felt the need to formalise the internal roles, defining rights, duties and election procedures, so that we could have a single text containing all the information about internal conduct. I wrote an internal regulation in order to reduce the imbalance in the workload, to promote inclusiveness and to relieve the members who were overloaded.

The introduction of this regulation was a key step to guarantee a correct distribution of tasks and the future stability of the association.

2.6 Second recruitment

What has been described so far refers to 2024 and ends with the winter exam session. As expected, the exam months brought a strong slowdown, because the priority was given to the exams.

According to our regulation, however, the end of the exam session and the beginning of the new period coincided with the second recruitment.

In those weeks I updated the presentations and introduced Aurora Rocketry to the first and second year students of Aerospace Engineering in Forlì and of Computer Science in Cesena. The choice of going to Cesena answered the need of the Electronics and Software departments to recruit students with previous experience.

In addition, I designed the whole flyer that represented Aurora Rocketry during this recruitment phase, shown in Figs. 6 and 7 in Appendix A.

The second recruitment was much more demanding than the first one, because it involved a real selection process. Each Department Head analysed the applications related to their own area, selected the most promising ones and held oral interviews to assess previous skills and motivation, in order to confirm or reconsider the first impressions. In case of a positive outcome, the candidate was officially admitted.

From that moment on, I also worked on graphics and multimedia material to represent the association at various events, as described in the dedicated sections.

2.7 Funding from LaBCC

The week of the second recruitment overlapped with an important event: a formal meeting with Silvia Cappelli, manager of the Forlì branch of La BCC Ravennate Forlivese Imolese, to discuss possible funding.

I also took part in the meeting, and I prepared a short presentation document about the association, describing the ongoing projects and our financial needs. The document was given to Cappelli as a reference for the committee in charge of granting the contributions.

It was our first formal meeting with an organisation outside the university, in which Aurora Rocketry presented the project to a potential sponsor and offered visibility in return, through social media channels, merchandising and other promotional means. It was a symbolic moment: exactly what we had been aiming at from the beginning.

On the same occasion, Cappelli offered to put us in contact with Monica Piraccini, an administrative officer of the Municipality of Forlì, in order to evaluate our participation in the MAE project, promoted by the Municipality. I took part in this meeting as well, but it did not lead to tangible results, and the application to the MAE project was temporarily set aside.

2.8 Application and admission to EuRoC 2025

At the beginning of April 2025 the applications for EuRoC 2025 opened. The short deadline took the association by surprise: we had to describe the prototype we intended to present, including its design solutions, design philosophy, state of progress and technical and organisational aspects.

At that moment Aurora Rocketry was focused on *Borealis*, which was not compatible with EuRoC. The project intended for the competition was still at an early stage and was not sufficient to meet the requirements of the application. This caused a general alarm and forced a drastic change of priorities: leaving *Borealis* aside and developing a new prototype that would represent the team at EuRoC 2025. This is how *Nemesis* was born.

We discussed the design method and, given the urgency, we decided to reuse the layout of *Borealis*, oversizing it in order to reach the altitudes required by the competition. The application was managed efficiently through a fair distribution of tasks: the questions of the form were assigned to the departments in charge, which completed them quickly. I was responsible for writing the motivation letter.

The collective effort led to the expected result: the official admission to EuRoC 2025.

From that moment on, new responsibilities appeared. While the departments took care of the technical part, I worked together with the Media & Logistics department on the production, editing and finishing of the official presentation video of Aurora Rocketry, which was required by EuRoC and published both on the channels of the association and on those of the competition.

Later on, several technical reports had to be produced: the *Concept Report*, the *Design Report* and above all the *Technical Report*, the most complete one, meant to describe the final project in every detail, including reliability checks, test reports and operational checklists to be filled in on site. I contributed to the *Technical Report*, in particular by working on the aerostructural analysis and on the structural verification of *Nemesis*, as described later.

2.9 Participation in IRESS 2025

While the application for EuRoC 2025 was being completed, the preparations for *IRESS 2025* (Italian Rocket Engineering Student Summit) began. This initiative has been promoted by the *PoliTo Rocket Team* of Politecnico di Torino since 2023. Every year the event brings together the Italian teams active in the rocketry field at a hosting university, where each group presents its projects and the results achieved during the last year.

In 2024 Aurora Rocketry had taken part only as a spectator (in Padua); in 2025, instead, it attended as one of the only two Italian teams admitted to EuRoC 2025.

Together with the Media & Logistics department, I took care of the presentation video of Aurora Rocketry dedicated to the event, which is still visible on the social media channels of the hosting team of the 2025 edition, *SASA Aerospace* of the University of Rome “La Sapienza”.

2.10 Participation in the 6th edition of the CASSINI Hackathon

Aurora Rocketry was contacted by the company *Teseas*, local organiser of the CASSINI Hackathon on the theme “Space for Healthcare”, held in Bologna from 16 to 18 May 2025.

The event, promoted by the European Commission, involved students and innovators on the use of space technologies in the healthcare field. Aurora Rocketry took part as a promotional partner and used the occasion to introduce itself during the opening ceremony.

The role of partner included:

- the promotion of the hackathon on the social media channels of the association;
- the possibility of sending teams as competitors;
- the opportunity to present the association at the opening ceremony, provided that a university professor was present.

Although no member was interested in taking part as a competitor, I considered it useful to exploit the visibility offered by the event. I therefore proposed the participation of prof. Fabrizio Ponti and of the Project Manager of *Nemesis*, Lorenzo Piccinini, as representatives of Aurora Rocketry, and both of them accepted.

I took care of the continuous promotion of the event on social media, structured the presentation by adapting it to the healthcare theme, and asked prof. Ponti to include any projects consistent with the focus on health and well-being.

The group therefore took part in the opening ceremony, presented its projects and attracted the interest of many participants, collecting relevant contacts that may be useful for the future of the association.

2.11 Participation in the We Make Future 2025 fair

From 3 to 6 June 2025 Aurora Rocketry, together with other student associations of the Department of Industrial Engineering, took part as an exhibitor in the *We Make Future 2025* fair, presenting its projects.

Together with the President Nicolò Basso, I designed the roll-ups, which can be seen in Figs. 8 and 9 in Appendix A, and which helped to strengthen the image of the association.

I was also present at the stand during the last day of the event, contributing to the representation of the association in front of the public.

2.12 Organisation of the trip to EuRoC 2025

Together with other members of the Media & Logistics department, I worked on the organisation of the trip to Constância (PT) for our participation in EuRoC 2025. The main activities were:

- collection of the personal information of all members for the organisation of the journey;
- management of the expense reimbursements through the regional call, keeping in touch with the competent authorities;
- booking of flights, accommodation and material transport through a specialised agency;
- organisation of local transport in order to guarantee mobility in Portugal;
- definition of the deadlines for the participants regarding all organisational activities, and check of their compliance.

2.13 Other activities

Besides the specific activities related to limited periods, I also carried out continuous general administration work. This included the constant management of members joining and leaving the association, the monitoring of the payment of the membership fees, and the design of the website of the association, which is now online and available [here](#).

3 Impact of the experience and final remarks

This experience had a significant impact on my overall path. The workload was high and came with academic and personal sacrifices, but it turned out to be one of the most formative and rewarding experiences I have had. It was possible to start a real project that created value not only for me, but also for dozens of students who are now growing inside Aurora Rocketry.

The commitment was not limited to technical activities: a relevant part of the work concerned the management and coordination of the less engineering-related areas, which were necessary to keep the project running and to keep the original promise of offering everyone a real extracurricular experience.

From a technical point of view, the aerostructural analysis of Nemesis was the area in which I grew the most: setting up a beam model, transferring the aerodynamic loads from CFD, handling the safety criteria and validating the choices allowed me to turn theory into concrete practice.

The work produced tangible results, but it also showed that there is room for improvement, especially on the organisational and human side. The urgency related to EuRoC concentrated our efforts on the technical part, leaving inclusiveness and member participation in the background. Looking ahead, a more selective recruitment and a more balanced distribution of responsibilities will give every participant the opportunity to contribute in a real way.

The main weakness was the exclusion of too many members, not because they lacked willingness, but because the structure was not adequate. Recognising clearly what did not work is the starting point for improving in future experiences.

4 Appendix A

4.1 Flyers created for the second recruitment



Figure 1: Flyer, black version.



Figure 2: Flyer, white version.

4.2 Roll-ups created for the WMF fair

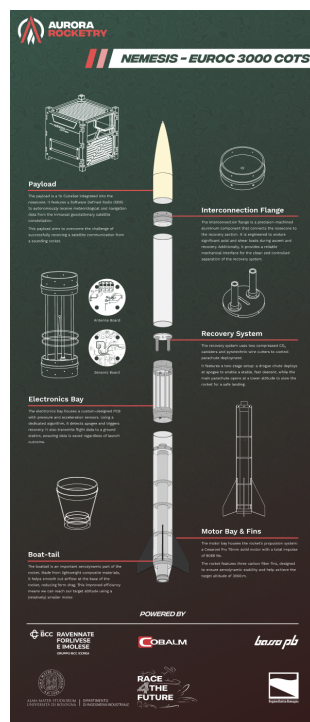


Figure 3: Nemesis roll-up.

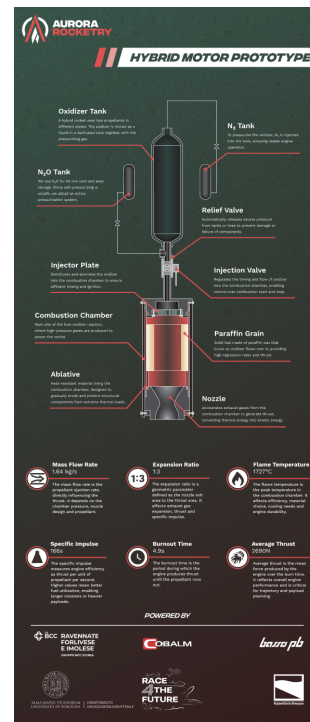


Figure 4: Hybrid motor roll-up.